

Transformation

- Motivation: We have a random variable X and we know its distribution. We are interested in a random variable Y which is some **transformation** of X , say $Y = g(X)$.
- Assume g is one-to-one:

$$p_Y(y) = \mathbb{P}[Y = y] = \mathbb{P}[g(X) = y] = \mathbb{P}[X = g^{-1}(y)] = p_X(g^{-1}(y)).$$

- Geometric Distribution: Recall that X was the flip number on which the first head appears. Let Y be the number of flips before the first head. Then $Y = X - 1$. In this case $g(x) = x - 1$.

$$p_Y(y) = p_X(x + 1) = \left(\frac{1}{2}\right)^{y+1}, \quad y = 0, 1, 2, \dots$$

- Let X be the random variable with a known pdf f_X . We are interested in the distribution of $Y = g(X)$.
- The CDF technique $\Rightarrow$ Work out the cdf of Y directly.
- Ex. Let $f_X(x) = 1/2$, $-1 < x < 1$, zero elsewhere, be the pdf of a random variable X . Let $Y = X^2$. Find the pdf of Y .

$$\mathbb{P}(Y \leq y) = \mathbb{P}(X^2 \leq y) = \mathbb{P}(-\sqrt{y} \leq X \leq \sqrt{y}) = \int_{-\sqrt{y}}^{\sqrt{y}} f_X(t) dt,$$

So,

$$F_Y(y) = \begin{cases} 0 & y < 0 \\ \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{2} dt = \sqrt{y} & 0 \leq y < 1 \\ 1 & y \geq 1 \end{cases}$$

Hence, the pdf is $f_Y(y) = \frac{1}{2\sqrt{y}}$, $0 < y < 1$, zero elsewhere.

- The transformation is one-to-one.
- Theorem. Let X be a continuous random variable with pdf $f_X(x)$ and support $\mathcal{S}_X$. Let $Y = g(X)$, where $g(x)$ is a one-to-one differentiable function, on the support of X . Denote the inverse of g by $x = g^{-1}(y)$ and let $dx/dy = d[g^{-1}(y)]/dy$. Then, the pdf of Y is given by

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|, \quad \text{for } y \in \mathcal{S}_Y,$$

where the support of Y is the set $\mathcal{S}_Y = \{y = g(x) : x \in \mathcal{S}_X\}$.

- Refer to $dx/dy = d[g^{-1}(y)]/dy$ as the **Jacobian** (denoted by J) of the transformation.

Example

- Let X has pdf

$$f_X(x) = \begin{cases} 1 & 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Find the pdf of $Y = -2 \log X$. The transformation $g(x) = -2 \log x$. The inverse of the transformation is $x = g^{-1}(y) = e^{-y/2}$. The Jacobian is

$$J = \frac{dx}{dy} = -\frac{1}{2}e^{-y/2}.$$

So the pdf of Y is

$$f_Y(y) = \begin{cases} f_X(e^{-y/2})|J| = \frac{1}{2}e^{-y/2} & 0 < y < \infty \\ 0 & \text{elsewhere} \end{cases}$$

Expectations

- Expectation: Let X be a random variable. If X is a continuous random variable with pdf $f(x)$ and $\int |x|f(x)dx < \infty$, then the **expectation** of X is

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} xf(x)dx.$$

If X is a discrete random variable with pmf $p(x)$ and $\sum_x |x|p(x) < \infty$, then the **expectation** of X is

$$\mathbb{E}(X) = \sum_x xp(x).$$

- $\mathbb{E}(X)$ is sometimes called **mathematical expectation** or **expected value** of X .

- Theorem 1: Let X be a random variable and Let $Y = g(X)$ for some function g .
 - (a). Suppose X is continuous with pdf $f_X(x)$. If $\int_{-\infty}^{\infty} |g(x)|f_X(x) < \infty$, then the expectation of Y exists and it is given by

$$\mathbb{E}(Y) = \int_{-\infty}^{\infty} g(x)f_X(x)dx.$$

- (b). Suppose X is discrete with pmf $p_X(x)$. Suppose the support of X is S_X . If $\sum_{x \in S_X} |g(x)|p_X(x) < \infty$, then $E(Y)$ exists and

$$\mathbb{E}(Y) = \sum_{x \in S_X} g(x)p_X(x)$$

- Theorem 2: Let $g_1(X)$ and $g_2(X)$ be functions of a random variable X . Suppose the expectations of $g_1(X)$ and $g_2(X)$ exist. Then, for any constants k_1 and k_2 , the expectation of $k_1g_1(X) + k_2g_2(X)$ exists and it is given by

$$\mathbb{E}(k_1g_1(X) + k_2g_2(X)) = k_1\mathbb{E}(g_1(X)) + k_2\mathbb{E}(g_2(X)).$$

- If the support of X is $\{a_1, a_2, \dots\}$, it follows that

$$\mathbb{E}(X) = a_1 p(a_1) + a_2 p(a_2) + a_3 p(a_3) + \dots .$$

This sum is seen to be a “weighted average” of the values of $a_1, a_2, \dots$.

This suggests that we call $\mathbb{E}(X)$ the arithmetic mean of the values of X .

- **Mean:** Let X be a random variable whose expectation exists. The **mean value** μ of X is defined to be $\mu = \mathbb{E}(X)$.

- Let X be a discrete random variable with support $\{a_1, a_2, \dots\}$ and with pmf $p(x)$, then,

$$\begin{aligned}\mathbb{E}(X - \mu)^2 &= \sum_x (x - \mu)^2 p(x) \\ &= (a_1 - \mu)^2 p(a_1) + (a_2 - \mu)^2 p(a_2) + \dots\end{aligned}$$

This sum may be interpreted as a “weighted average” of the squares of the deviations of the numbers $a_1, a_2, \dots$ from the mean values μ .

- Variance: Let X be a random variable with finite mean μ and such that $\mathbb{E}[(x - \mu)^2]$ is finite. Then the **variance** of X is defined to be $E[(X - \mu)^2]$. It is usually denoted by σ^2 or by $\text{Var}(X)$.

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$$\sigma^2 = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \mathbb{E}(X^2) - \mu^2.$$

- σ (positive square root of the variance) is called **standard deviation** of X . The number σ is sometimes interpreted as a measure of the dispersion of the points of the space relative to the mean value μ . If the space contains only one points k for which $p(k) = 1$, then $\mu = k$ and $\sigma = 0$.

Moment Generating Function (mgf)

- mgf: $M(t) = \mathbb{E}(e^{tX})$ for some random variable X . $M(0) = 1$.

- Continuous:

$$M(t) = \int e^{tx} f(x) dx.$$

- Discrete:

$$M(t) = \sum_i e^{ta_i} p(a_i).$$

- Theorem: $X \stackrel{D}{=} Y$ iff $F_X(z) = F_Y(z)$ iff $M_X(t) = M_Y(t)$.

- $M'(0) = \mathbb{E}(X) = \mu$

- Continuous:

$$\begin{aligned} M'(t) &= \frac{dM(t)}{dt} = \frac{d}{dt} \int_{-\infty}^{\infty} e^{tx} f(x) dx = \int_{-\infty}^{\infty} \frac{d}{dt} e^{tx} f(x) dx \\ &= \int_{-\infty}^{\infty} x e^{tx} f(x) dx. \end{aligned}$$

- Discrete:

$$M'(t) = \frac{dM(t)}{dt} = \sum_x x e^{tx} p(x)$$

- $M''(0) = \mathbb{E}(X^2)$.
- $M^{(m)}(0) = \mathbb{E}(X^m)$, $m = 1, 2, 3, \dots$

References

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- [4] *John E. Freund's Mathematical Statistics with Applications* by Irwin Miller Marylees Miller